Midterm 1

Name: Solution

Student ID: ______________________

By signing below, you affirm that you have neither given nor received unauthorized help on this exam.

Signature: ______________________

DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO! YOU WILL GET A ZERO IF YOU DO.

This exam contains 6 pages (including this cover page) and 4 problems. The exam will be graded out of 60 points. **This is an individual exam.** You are not allowed to use a calculator, any additional notes, look at anyone else's paper, have extra paper, talk to anyone, consult the internet, anything. It is considered cheating if you use additional notes, have wandering eyes, are caught talking, etc. Absolutely no headphones, smartwatches, or cellphones out! **I have a zero tolerance policy for cheating.**

Instructions: Print your name and student ID number above AND ON EVERY PAGE. You may use any available space on the exam for scratch work.

Good Luck!

To Receive Full Credit

- **Answer must be written in the box provided!** Show your work on the rest of the page.
- You must **show your work** to receive credit.
- **Proper Notation and Organized.** Clear, concise, and correct use of mathematical notation is part of a correct answer. Work should be organized in a reasonable neat and coherent way in the space provided. Work scattered all over the page without a clearing ordering is not acceptable.

Do not write in the table to the right.

Page	Points	Score
1	13	
2	8	
3	13	
4	10	
5	16	
Total:	60	

1. (8 points) Find the general solution for $y^{[5]}(x) - 6y^{[4]}(x) + 13y^{[3]}(x) = 0$

$$\boxed{y(x) = c_1 + c_2 x + c_3 x^3 + c_4 e^{3x} \cos(2x) + c_5 e^{3x} \sin(2x)}$$

$r^5 - 6r^4 + 13r^3 = 0$

$r^3(r^2 - 6r + 13) = 0$

$r = 0$ mult 3, $r = 3 \pm 2i$

$r = \dfrac{6 \pm \sqrt{36 - 4(13)}}{2}$

$= \frac{6}{2} \pm \frac{1}{2}\sqrt{-16}$

$= 3 \pm 2i$

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2. Find the Laplace transform of the following functions, if it exists. If it does not exist, justify. Your answer should not have any derivatives and/or integrals left in them.

(a) (5 points) $\mathcal{L}\left[e^{2x}\cos(5x)\right]$.

$$\boxed{\frac{(p-2)}{(p-2)^2 + 25}}$$

(b) (8 points) $\mathcal{L}\left[f(x)\right]$, where $f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ x-1 & 1 \leq x < 3 \\ 2 & x \geq 3 \end{cases}$

$$\boxed{\frac{1}{p^2}\left(e^{-p} - e^{-3p}\right)}$$

By Integral Definition or...

$$f(x) = (x-1)\mathcal{U}(x-1) - (x-1)\mathcal{U}(x-3) + 2\mathcal{U}(x-3)$$
$$= (x-1)\mathcal{U}(x-1) - (x-3)\mathcal{U}(x-3)$$
$$\mathcal{L}[f(x)] = \mathcal{L}[(x-1)\mathcal{U}(x-1)] - \mathcal{L}[(x-3)\mathcal{U}(x-3)]$$
$$= \frac{e^{-p}}{p^2} - \frac{e^{-3p}}{p^2}$$

3. Find the inverse Laplace transform of the following functions, if it exists. If the answer involves a convolution, you need to compute the integral. If it does not exist, justify.

(a) (5 points) $\mathcal{L}^{-1}\left[\dfrac{4e^{-3p}}{p^2+16}\right]$

$$\boxed{\mathcal{U}(x-3)\sin(4(x-3))}$$

$$\mathcal{L}^{-1}\left[e^{-3p}\left(\frac{4}{p^2+4^2}\right)\right]$$

(b) (8 points) $\mathcal{L}^{-1}\left[\dfrac{2p+5}{p^2+4p+5}\right]$

$$\boxed{2e^{-2x}\cos x + e^{-2x}\sin x}$$

$$\frac{2p+5}{p^2+4p+5} = \frac{2(p+2)+1}{(p+2)^2+1}$$

$$= \frac{2(p+2)}{(p+2)^2+1} + \frac{1}{(p+2)^2+1}$$

(c) (10 points) $\mathcal{L}^{-1}\left[\dfrac{5p+12}{p(p^2+4)}\right]$

$$\boxed{3+\frac{5}{2}\sin(2x)-3\cos(2x)}$$

$$\frac{5p+12}{p(p^2+4)}=\frac{5}{p^2+4}+\frac{12}{p(p^2+4)}$$

$$=\frac{5}{p^2+4}+\frac{3}{p}-\frac{3p}{p^2+2^2}$$

PFD $\longrightarrow$ rather than computing convolution integral

$$\frac{12}{p(p^2+4)}=\frac{A}{p}+\frac{Bp+C}{p^2+4}$$

$$Ap^2+4A+Bp^2+Cp=12$$

$$\begin{cases} A+B=0 \\ C=0 \\ 4A=12 \end{cases} \Rightarrow \begin{matrix} A=3 \\ B=-3 \end{matrix}$$

4. (16 points) Solve via Laplace Transform $\begin{cases} y'' + y = xe^{-x} \\ y(0) = 0, \quad y'(0) = 1 \end{cases}$.

Should you determine convolution necessary, you must compute the integration.

$$\boxed{y(x) = \sin x - \frac{1}{2}\cos x + \frac{1}{2}e^{-x} + \frac{1}{2}xe^{-x}}$$

Let $\mathcal{Y} = \mathcal{L}[y(x)]$

$$p^2\mathcal{Y} - py(0) - y'(0) + \mathcal{Y} = -\frac{d}{dp}\left(\frac{1}{p+1}\right)$$

$$p^2\mathcal{Y} - 1 + \mathcal{Y} = \frac{1}{(p+1)^2}$$

$$\mathcal{Y} = \frac{1}{p^2+1} + \frac{1}{(p+1)^2(p^2+1)}$$

If you do convolution

$$\mathcal{L}^{-1}\left[\left(\underset{\uparrow \ \sin x}{\frac{1}{p^2+1}}\right)\left(\underset{\uparrow \ xe^{-x}}{\frac{1}{(p+1)^2}}\right)\right]$$

$$\frac{1}{(p+1)^2(p^2+1)} = \frac{Ap+B}{p^2+1} + \frac{C}{p+1} + \frac{D}{(p+1)^2}$$

$$\mathcal{Y} = -\frac{1}{2}\left(\frac{p}{p^2+1}\right) + \frac{1}{2}\left(\frac{1}{p+1}\right) + \frac{1}{2}\left(\frac{1}{(p+1)^2}\right) + \frac{1}{p^2+1}$$

Math 135: Laplace Transforms

$\boldsymbol{f(x)}$	$\boldsymbol{\mathcal{L}[f(x)] = F(p)}$
$f(x)$	$\int_0^\infty e^{-px} f(x) dx$
$f^{[n]}(x)$	$p^n F(p) - p^{(n-1)} f(0) - \cdots - f^{[n-1]}(0)$
1	$\frac{1}{p}$
x^n	$\frac{n!}{p^{n+1}}$
e^{ax}	$\frac{1}{p-a}$
$\sin(ax)$	$\frac{a}{p^2+a^2}$
$\cos(ax)$	$\frac{p}{p^2+a^2}$
$\sinh(ax)$	$\frac{a}{p^2-a^2}$
$\cosh(ax)$	$\frac{p}{p^2-a^2}$
$e^{ax} f(x)$	$F(p-a)$
$x^n f(x)$	$(-1)^n \frac{d^n F(p)}{dp^n}$
$\mathcal{U}(x-a)$	$\frac{e^{-ap}}{p}$
$f(x-a)\mathcal{U}(x-a)$	$e^{-ap} F(p)$
$\delta(x)$	1
$\delta(x-x_0)$	e^{-px_0}
$f(x) * f(x) = \int_0^x f(t) g(x-t) dt$	$F(p)G(p)$